Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
(Approved by AICTE New Delhi & Govt. of Maharashtra, Affiliated to University of Mumbai)
(Religious Jain Minority)

Department of Computer Science & Engineering (AI & ML)

# An Intelligent Library Management System with book recommendation

**Theme-** SDG Goals

SDG 4 – Quality Education

SDG 9 – Industry, Innovation and Infrastructure

**Category**- Software

**Group No -** 2

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# Outline

- Introduction
- Literature Survey
- Limitations of  existing systems
- Problem statement and Objectives
- Proposed Methodology
- Experimental Setup
- Implementation
- Results & Analysis
- Conclusion and Future work
- References

# 1 Introduction

## Background of the Topic:

- Libraries are vital knowledge hubs in educational institutions.
- Book searching, issuing, and tracking processes are inefficient.
- Limited automation reduces accessibility and overall user efficiency

## Importance of the Problem:

- Students waste significant time searching for books manually.
- Lack of intelligent recommendation reduces learning efficiency.
- Errors in manual record-keeping affect transparency and trust.
- Improving library systems directly enhances quality of education and equal access to knowledge.

# Introduction

## Current Challenges:

- Manual or slow book search process.
- No personalized recommendation system.
- Unclear real-time availability of books.
- Human errors in record maintenance.

## Motivation (Why We Chose This Problem):

- Observed students standing in queues and struggling to find required books.
- Lack of transparency between librarian records and student access.
- Need for a smarter, faster, and fair system in modern educational environments.

## Objectives :

- Develop a chatbot-based smart search system for instant book discovery.
- Provide intelligent book recommendations based on topics/keywords.
- Enable real-time availability tracking of library resources.
- Ensure fair allocation of popular books using slot/queue mechanisms.

# Introduction

## Why we Chose AGENTIC-AI ?

- Students don't always know exact book titles.
- They need guidance based on level and interest.
- Enhances user experience without modifying core system.
- AI asks clarifying questions.
- Uses backend APIs as tools.
- Suggests books from database only.

## How It Works (Architecture + Safety)

Student Query
↓
AI analyzes intent
↓
(Optional) Asks clarification
↓
Calls backend search API
↓
Receives database results
↓
Generates short explanations

# 2 Literature Survey

## Survey of Existing Solutions

- Traditional Library Management Software (manual + barcode based systems)
- Digital Library Portals (OPAC systems)
- Book Recommendation Websites

## Identified Limitations

- No AI-based chatbot or natural language search
- Lack of personalized book recommendations
- No real-time transparency for book availability
- Manual dependency on librarian for queries

## Techniques Commonly Used

- Barcode / RFID Systems – for tracking books
- Database Management Systems (MySQL, Oracle) – storing records
- Rule-Based Recommendation – limited suggestions

# Literature Survey

| Author & Year | Technique / Method Used | Dataset Used | Key Results | Limitations |
|---|---|---|---|---|
| Sharma et al., 2020 | Barcode-Based LMS | College Library Records | Improved issue/return speed | No intelligent search or recommendation |
| Patel & Mehta, 2021 | RFID Tracking System | University Library Data | Reduced manual errors | High cost, no personalization |
| Kumar et al., 2022 | Keyword Search Digital Portal | Online Book Database | Easy online catalog access | No real-time fairness or chatbot |
| Singh & Roy, 2023 | Rule-Based Recommendation | Student Borrow History | Suggested related books | Limited accuracy, not AI-driven |

# 3 Limitations of Existing Systems

**Existing System:** koha LIBRARY SYSTEM

**KOHA Limitations:**

- No chatbot interaction
- No intelligent book recommendation
- Admin based system (Manual entry required)
- Complex UI for users
- Technical setup required

AI chatbot where students can ask for books by topic and get instant suggestions.

Both students and admin can access books

Simple and user-friendly interface designed for easy student use.

# 4 Problem statement and Objectives

Existing library systems lack smart search and personalized book recommendations, **making it hard for students** to quickly find relevant books and issue them easily.

## OBJECTIVES OF THE PROPOSED SYSTEM

-To provide smart book recommendations.

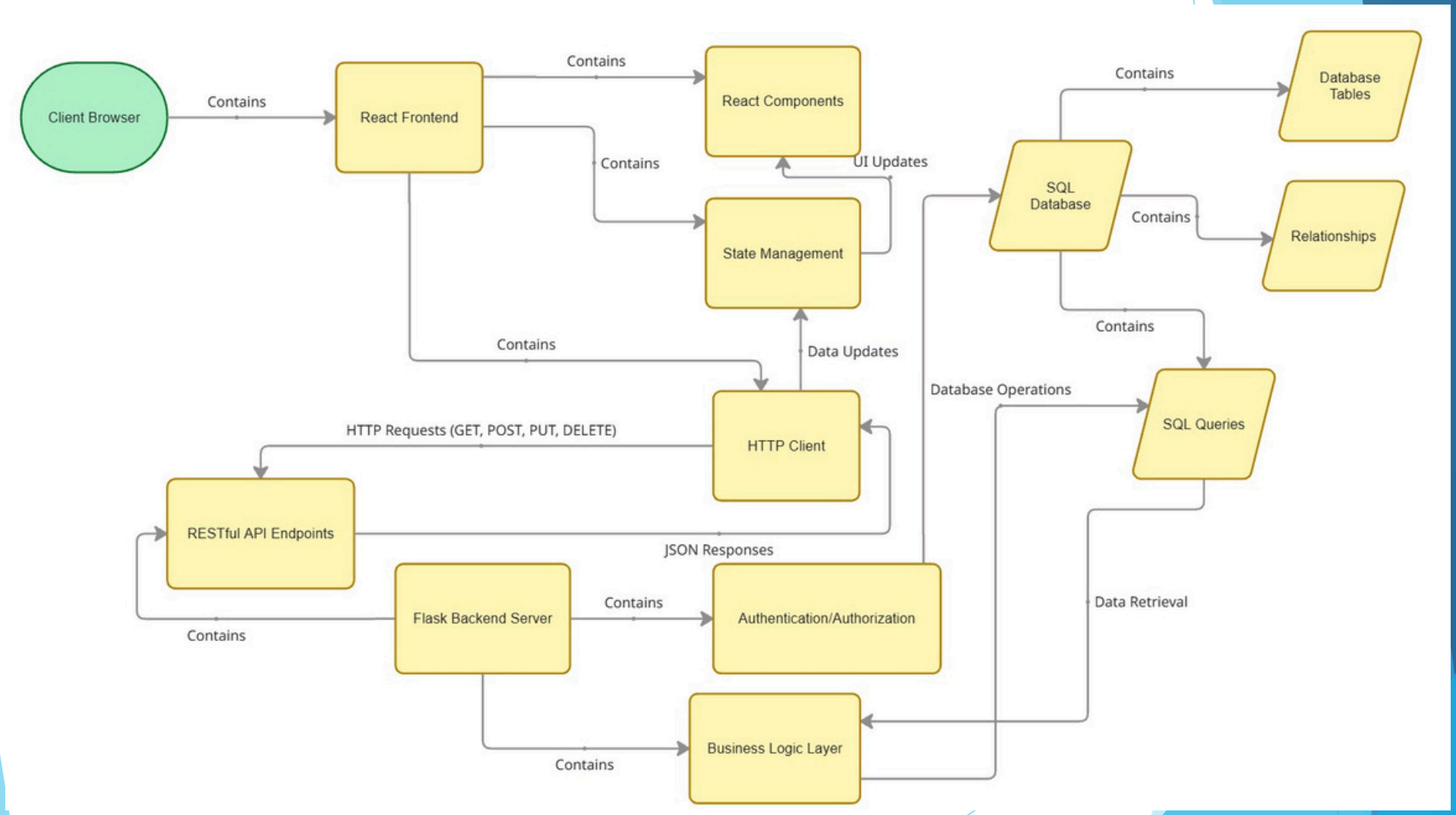
-To help students quickly find relevant books.

-To allow both students and admin to access the system.

-To design a user-friendly interface for easy navigation.

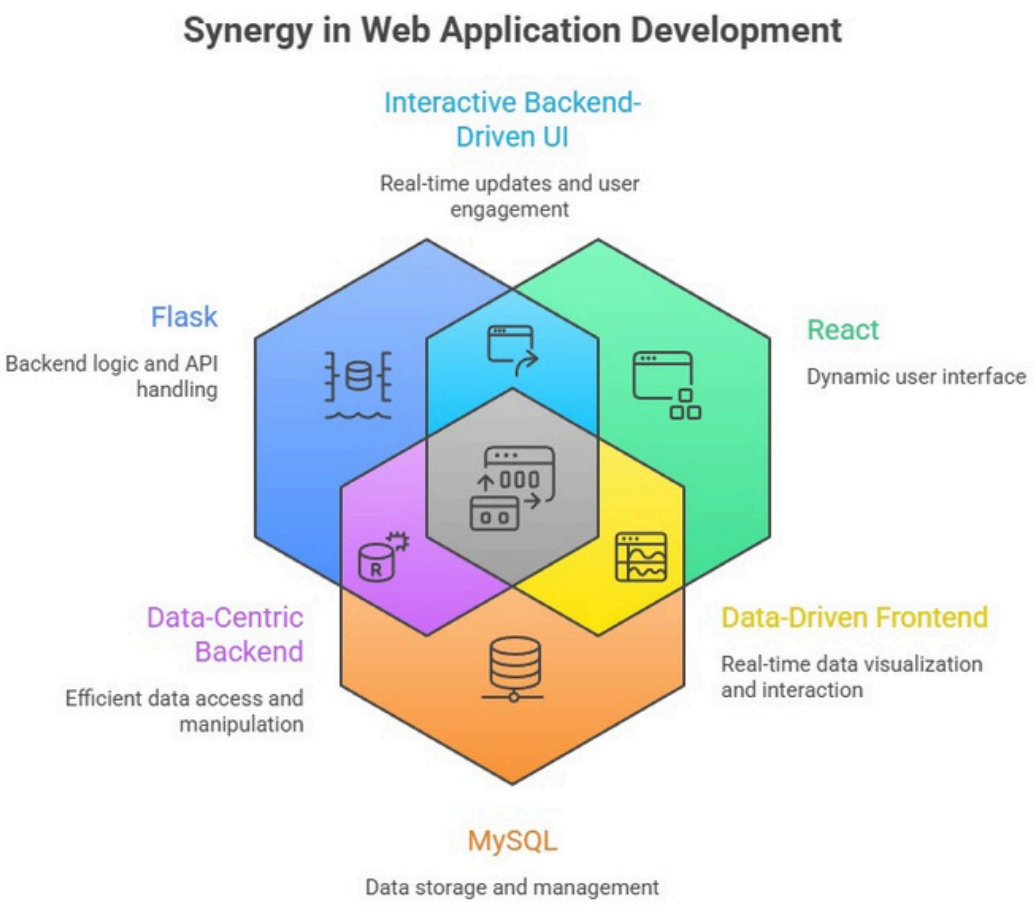
-To show real-time book availability status to users.

# 5 Proposed Methodology



# 6 Experimental Setup

## TechStack:



# 9 Conclusion and Future work

- Successfully developed a Smart Library Management System with chatbot-based search and automated tracking.
- Enabled faster book discovery, real-time availability checking, and intelligent recommendations.
- Reduced manual effort for librarians and improved overall user experience for students.
- Demonstrated improved efficiency and transparency compared to traditional systems.

## Future Work:

- Integration with mobile application for on-the-go access.
- Advanced machine learning models for more accurate recommendations.
- Voice assistant support for accessibility.
- Multi-library / cloud integration for wider resource sharing.
- Enhanced data analytics dashboard for administrators.

# 10 References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# Thank You…!!